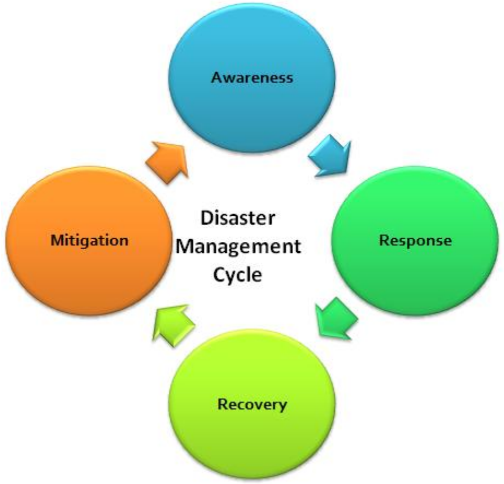
Disaster Management Cycle

Fall 2026

Learning Outcomes

After this lecture, the students will be able to;

- **Know** – about the disaster risk management phases
- **Identify** - suitable hazard risk mitigation and preparedness approaches and disaster response and recovery measures



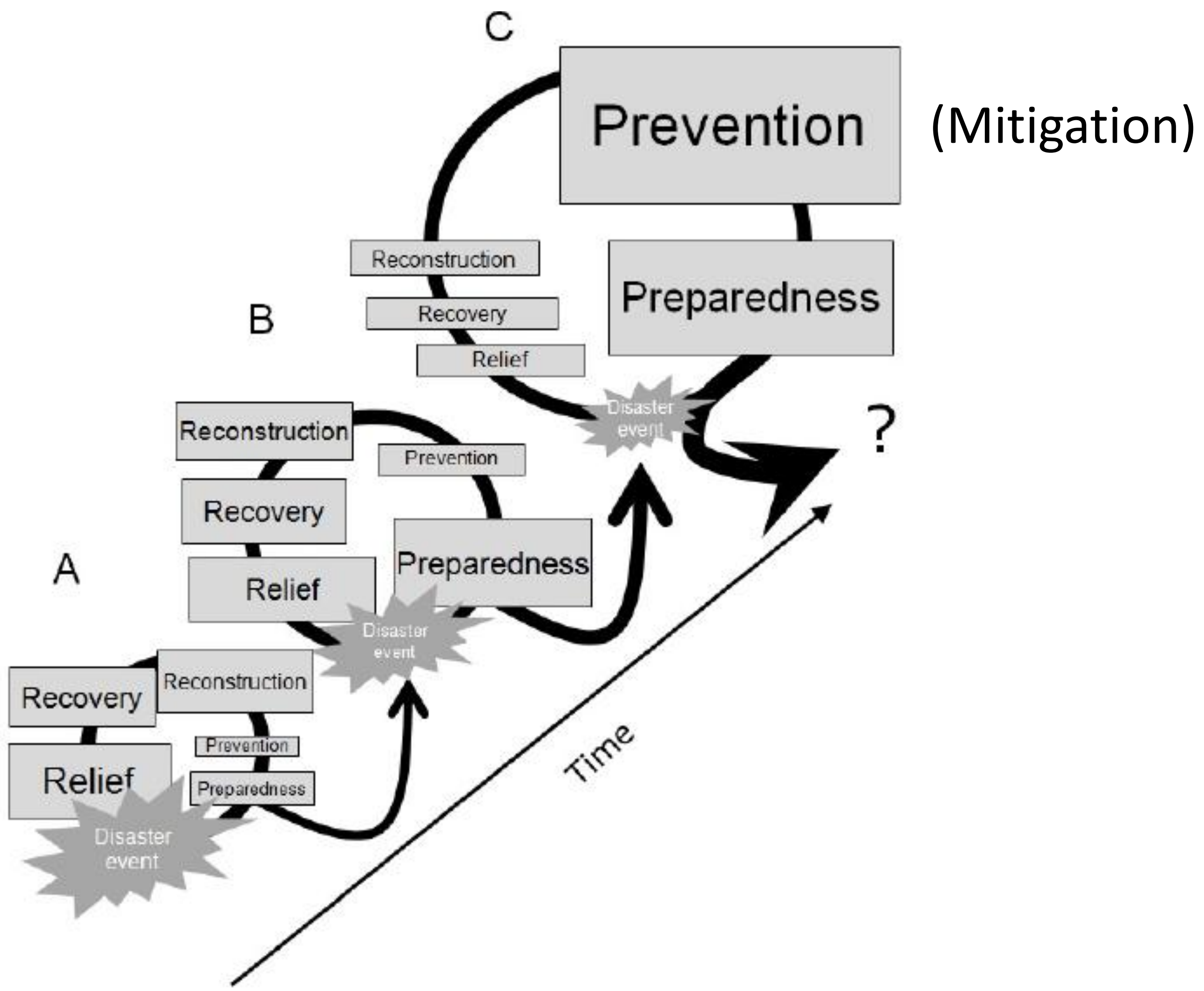
Disaster Mitigation & Preparedness

Reading Material

- Introduction to Emergency Management By George Haddow, Jane Bullock, Damon P. Coppola
  - Chapters 3 & 6

DM Cycle

- It is not a series of events which start and stop with each disaster occurrence.
- Disaster and managing it is a continuum of interlinked activity.



Disaster cycle (rather spiral) and its development through time

Mitigation

- Attempt to prevent events from becoming disasters – How?
- Sustained action to reduce or eliminate the risks to people and property from hazards and their effects.
  - Class Discussion: What is meant by the term sustained actions?
- Measures taken to reduce <u>both the effect of the hazard and the vulnerable conditions</u> to it in order to reduce the scale of a future disaster
  - Examples?

Mitigation Activities

- Mitigation activities can be focused on
  - Hazard itself

  Example: water management in drought-prone areas.

  - Elements exposed to the threat

  Examples:

  - strengthening structures to reduce damage when a hazard occurs.
  - reducing the economic and social vulnerabilities of potential disasters.

Mitigation Alternatives

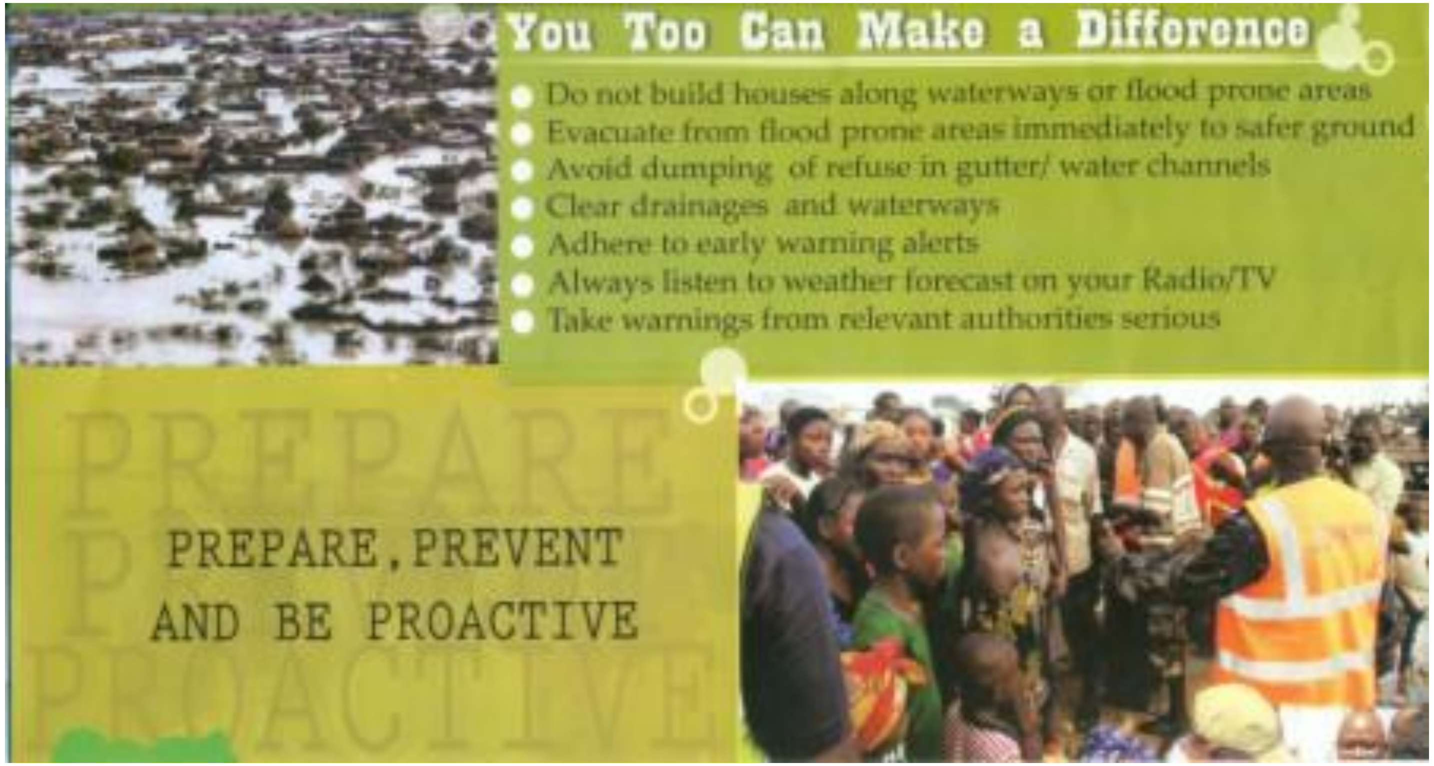
- Prevent the creation of the hazard in the first place.
- Change the nature or the size of the hazard.
- Separate the hazard from that which it might affect
- Modify the basic characteristics of a hazard.
- Research what others are doing

For Examples, see https://www.iaem.org/portals/25/documents/is1_Unit3.pdf

FEMA document

More Examples of Mitigations

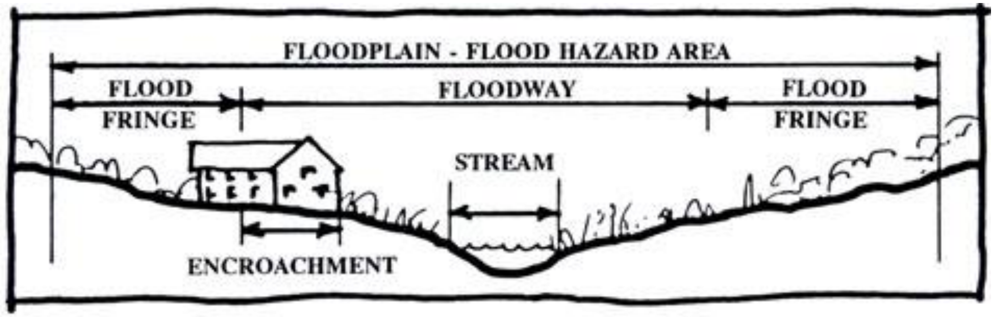
Public education and outreach activities designed to reduce loss of life and property destruction



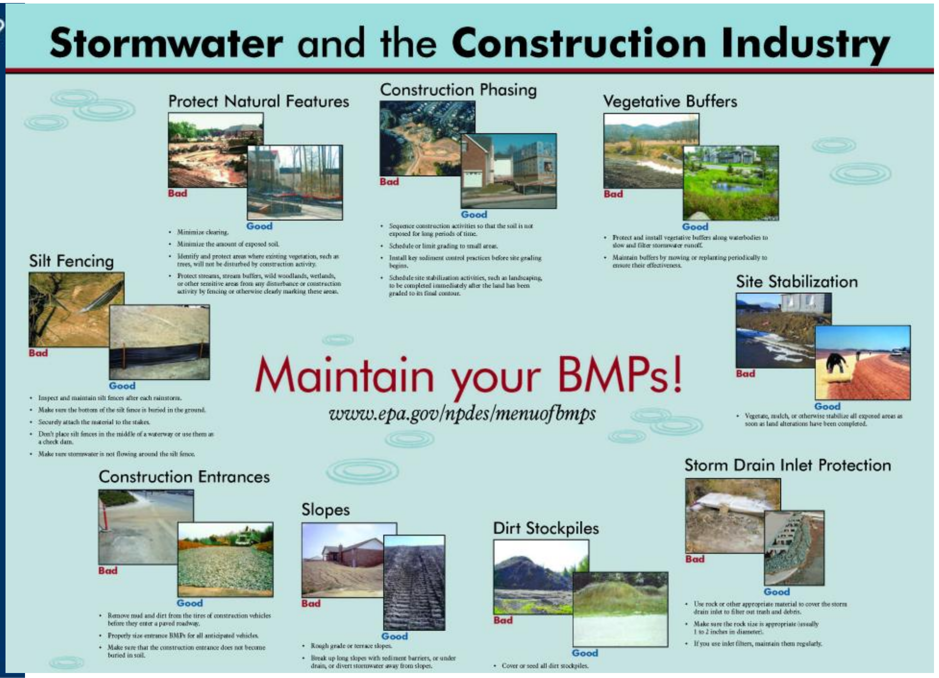
Examples conti…

FEMA document

- Floodplain management and land-use regulations



Source: Chester County Planning Commission, Rural Community Design Guide, 1995.



Examples conti…

FEMA document

Building codes, seismic design standards, and wind-bracing requirements for new construction, or repairing or retrofitting existing buildings

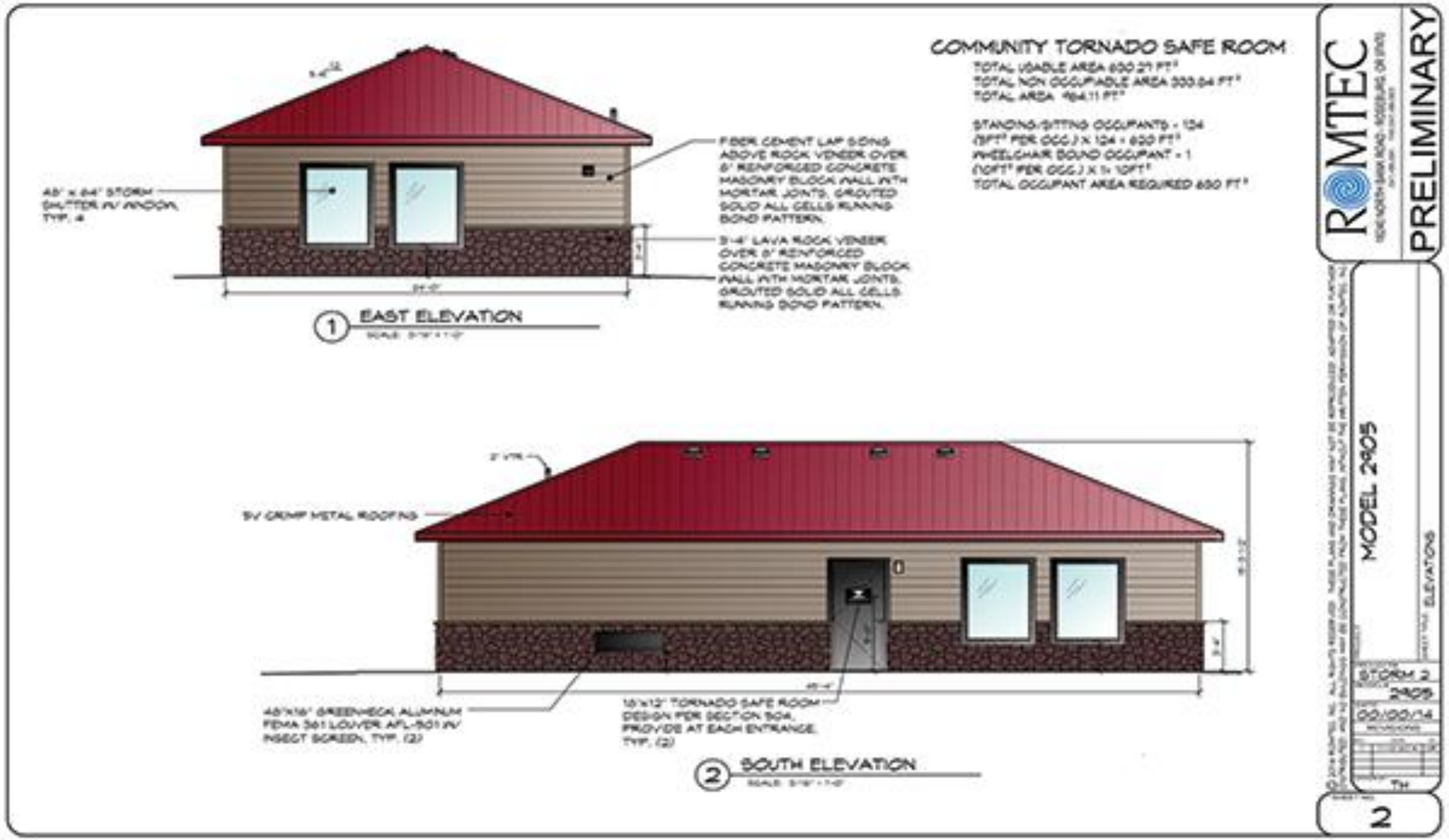
Examples conti…

FEMA document

Acquiring damaged homes or businesses in flood-prone areas, relocating the structures, and returning the property to open space, wetlands, or recreational uses

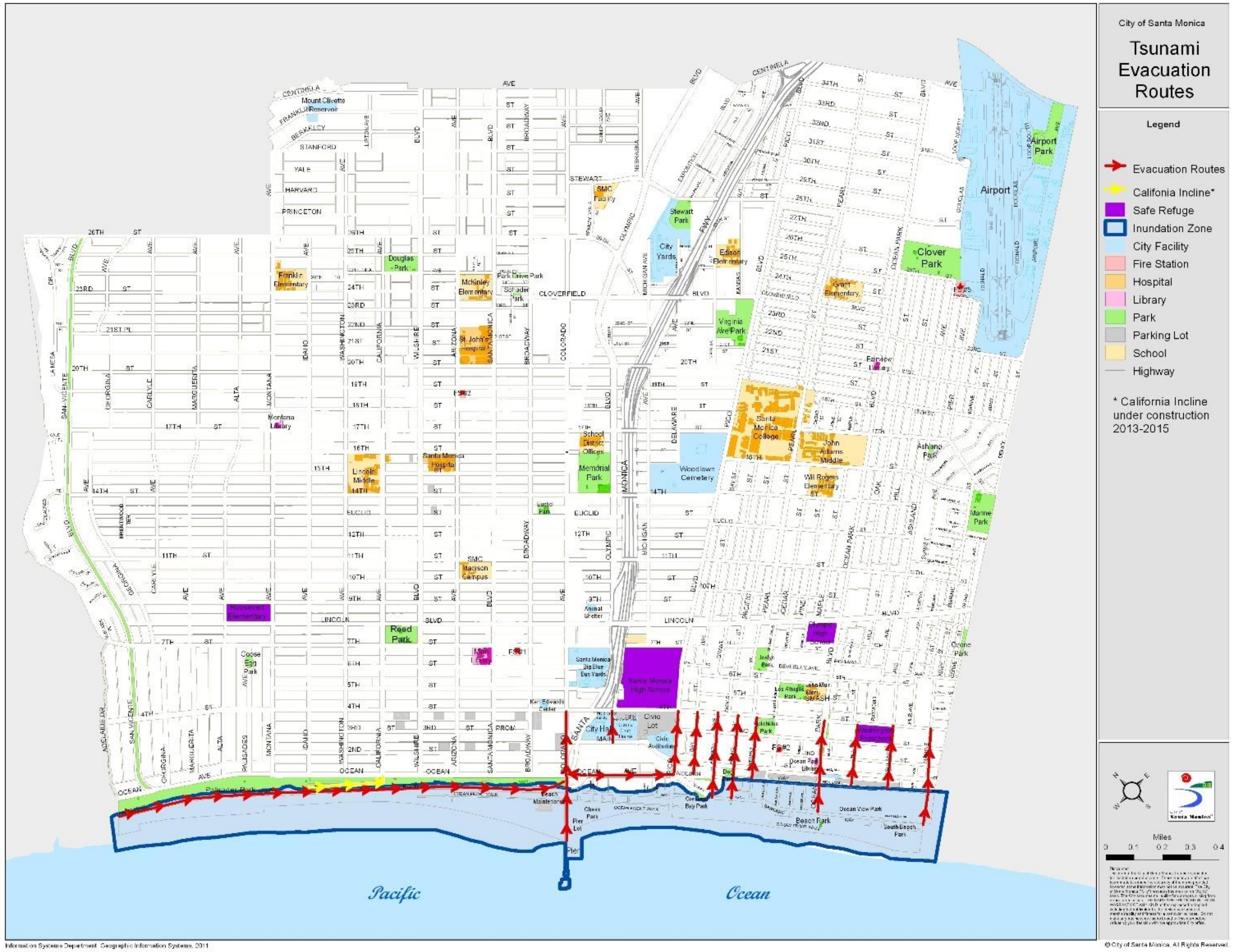
Structure Relocation: Most expensive but most effective Land use Planning tool

Identifying, and refurbishing shelters and safe rooms to help protect people in their homes, public buildings, and schools in hurricane- and tornado-prone areas



Examples of Mitigations: Conti…

Development of hazard-specific evacuation routes

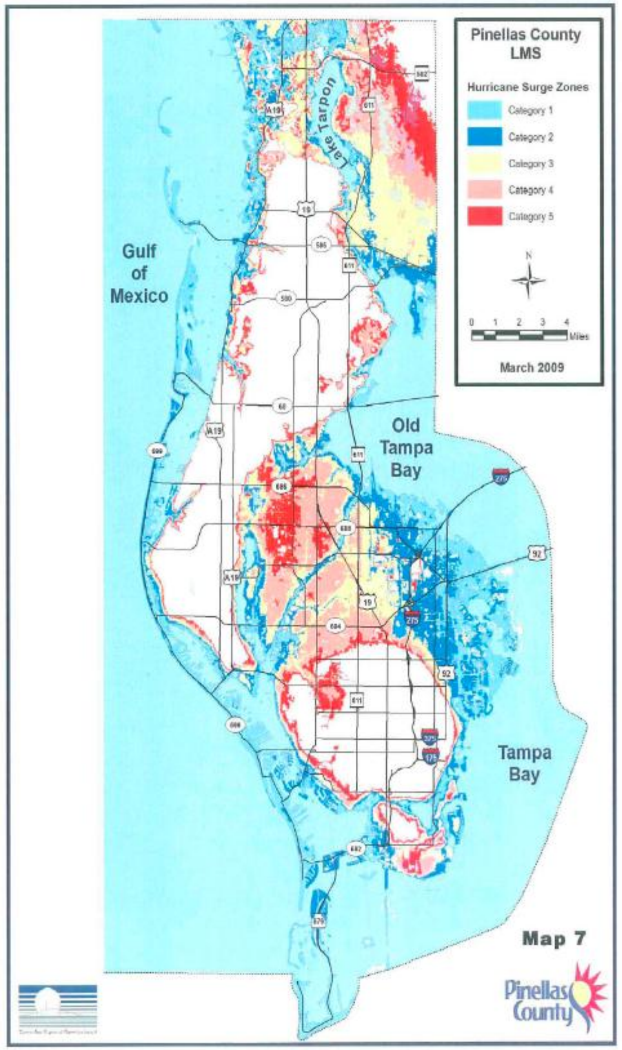


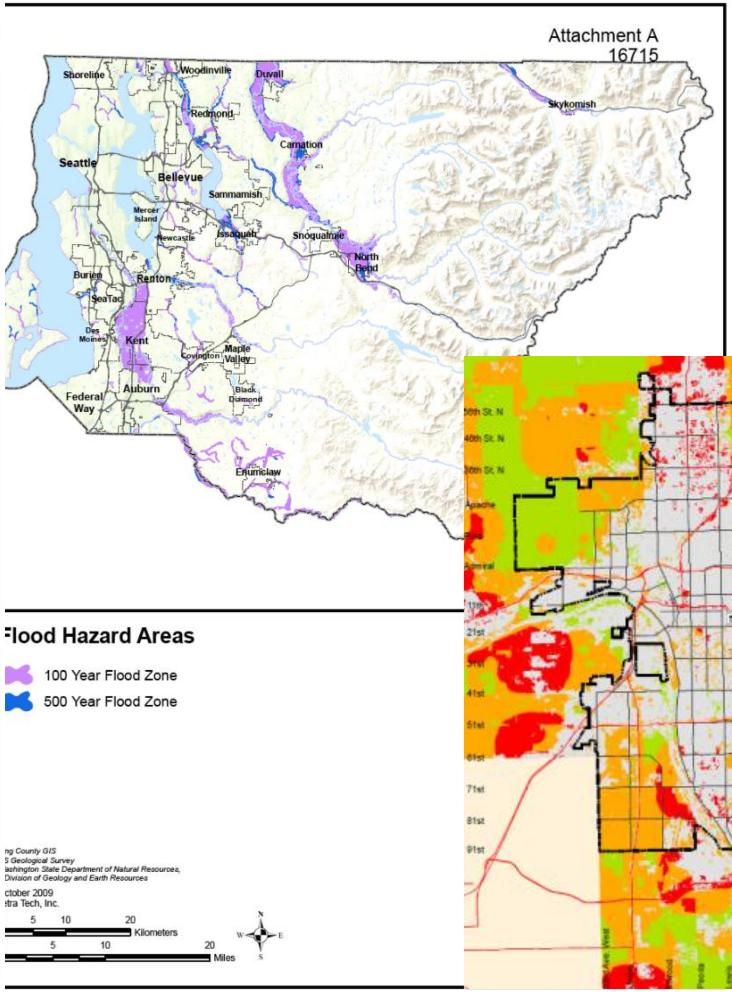
Flood Hazard Maps

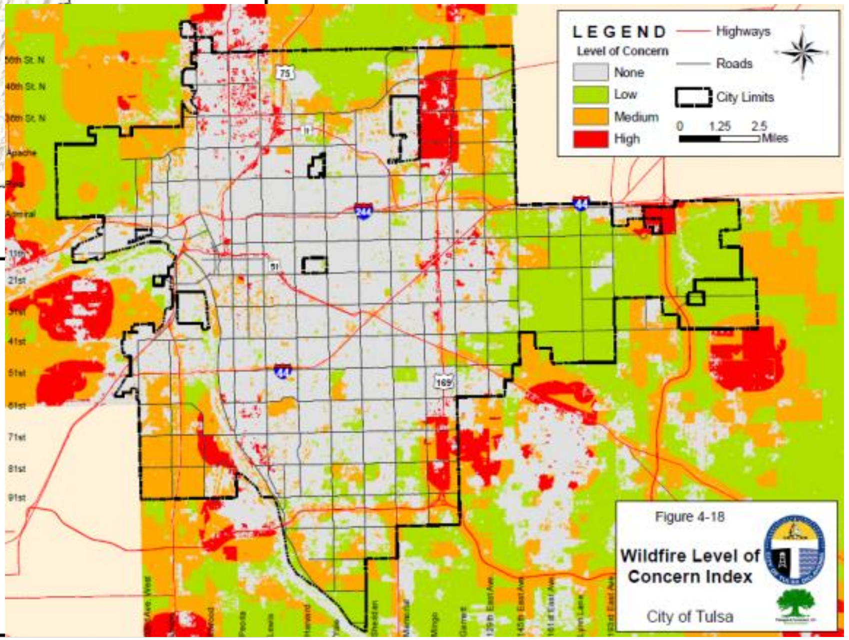
- Maps that graphically provide information on inundation as well as on evacuation in an easy-to-understand format
- Flood hazard mapping is an example of non-structural mitigation measures
- Assist local residents in becoming aware of the
  - vulnerability of the area they live in,
  - the important roles that they play in disaster prevention activities and
  - proper evacuation in the event of floods.

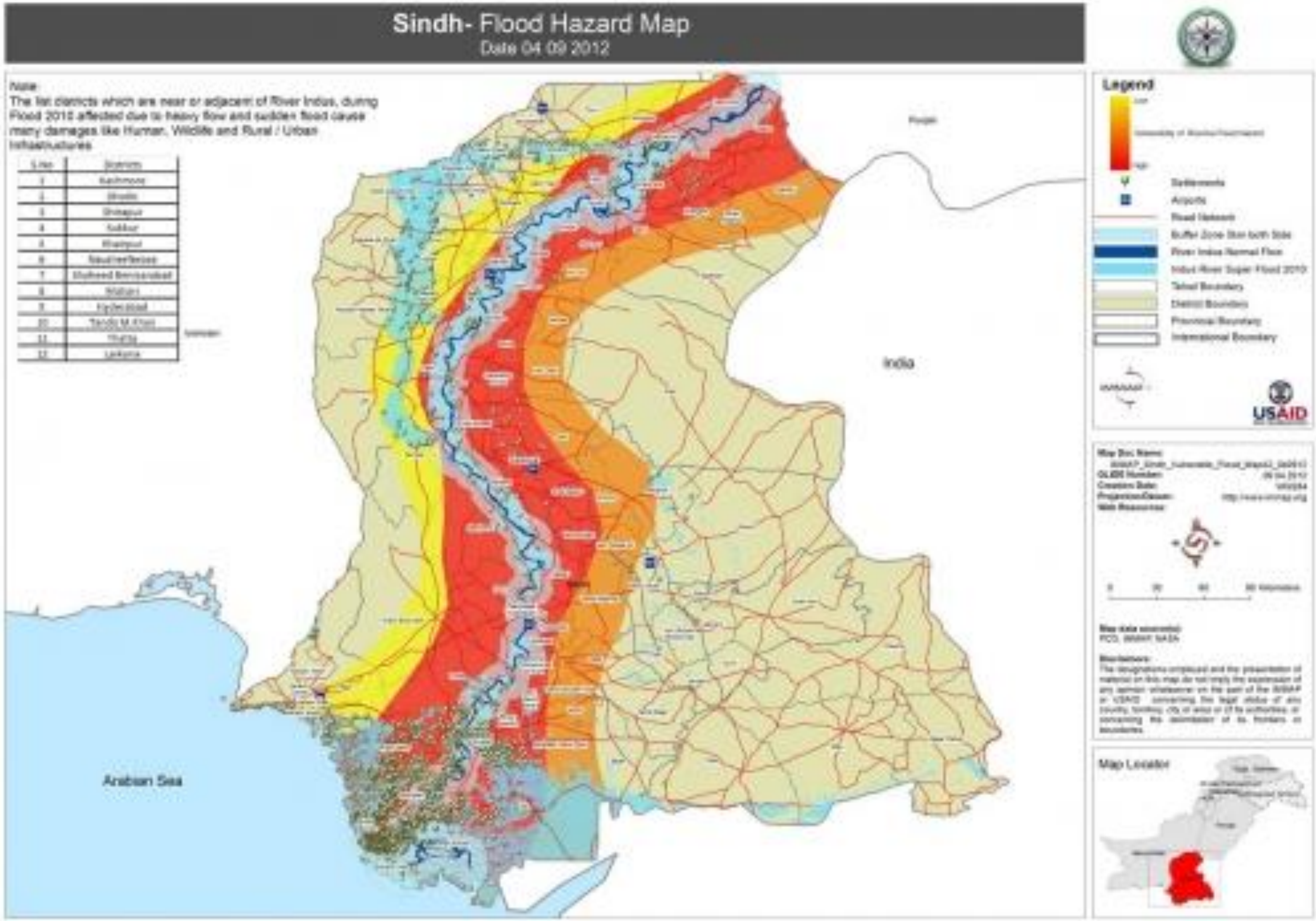
Examples of Mitigations: Conti…

Mapping of hazard or potential hazard zones, using geospatial techniques









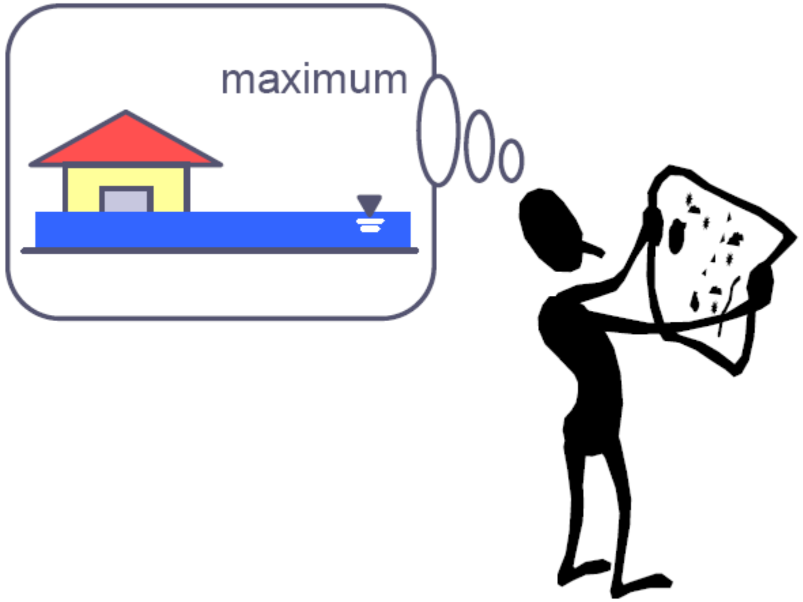
More about Flood Hazard Maps

How to Represent Information on Flood Hazard Maps?

- Should not convey a message that information shown on these maps is **unchanged**
- Predicted inundation depths are usually shown in different colors
- Flood danger level shown as the inundation depth
- levee-break spots
- Safe refuge and safe evacuation route
- Non-colored areas on the hazard maps may be mistaken as non-inundation areas, whereas these are merely results from the flood simulation
- In flash floods, how do you show the flow velocities on the map?

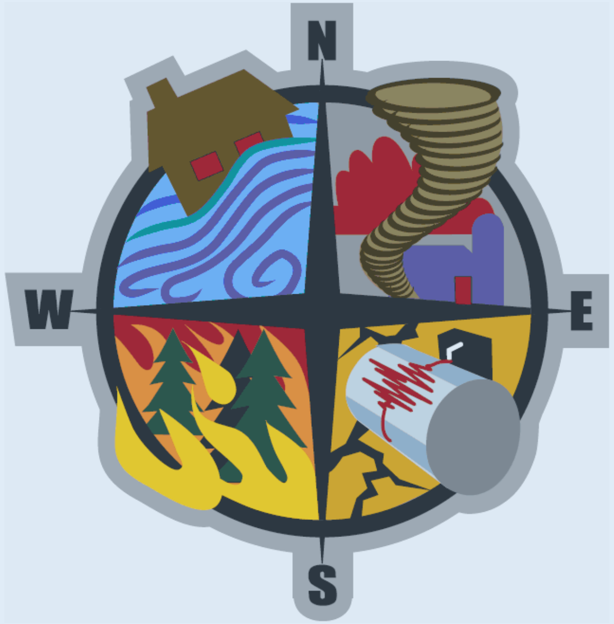
Hazard Maps

- **Have no value** without any educational activities
- Local residents should be encouraged to actively participate in the development of Flood Hazard Maps



Interesting Fact!

A study done for FEMA by the Multi-Hazard Mitigation Council (MMC) of the National Institute for Building Sciences (NIBS) found that <u>for every $1 invested in mitigation, $4 would be saved in future losses</u>



Preparedness

Disaster Planning and Preparation

Better to be prepared a year early than a day late

Disaster Preparedness

- A state of readiness to respond to a disaster.
- Protective process to respond rapidly to a disaster to cope with it.

- The capacity to respond and recover from emergencies and disasters is only developed through planning, training, and exercising —the heart of preparedness.
  - ✓ assess the exposure to risk
  - ✓ measures to build capacity and/or surge capacity

Whose Responsibility?

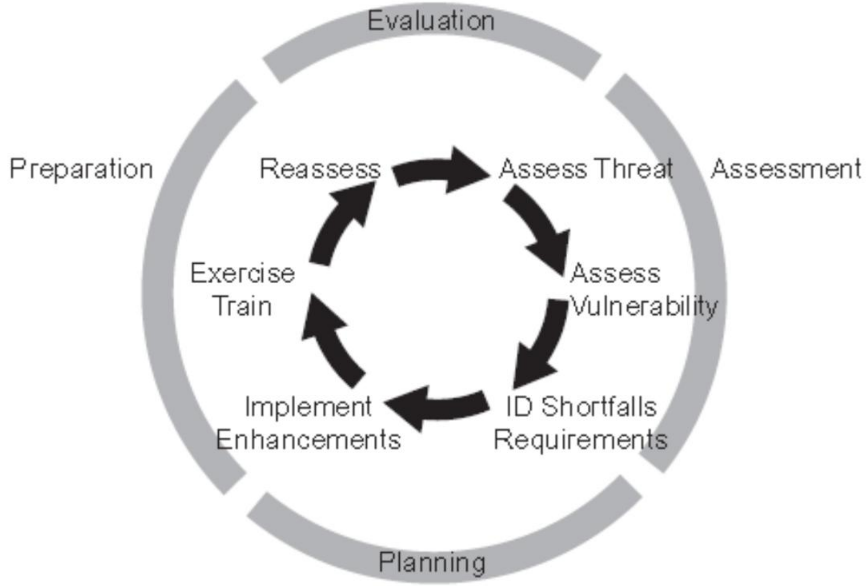
- Expands beyond the boundaries of emergency management agencies.
- Those susceptible to the consequences of disasters must ensure their preparedness.
  - All organizations: private, governmental, non-governmental

Scope?

- Focus not only on the protection of citizens, property, and essential government services but also on ensuring the viability and sustainability of the community — including its
  - businesses and markets,
  - social services and character

Preparedness: Activities

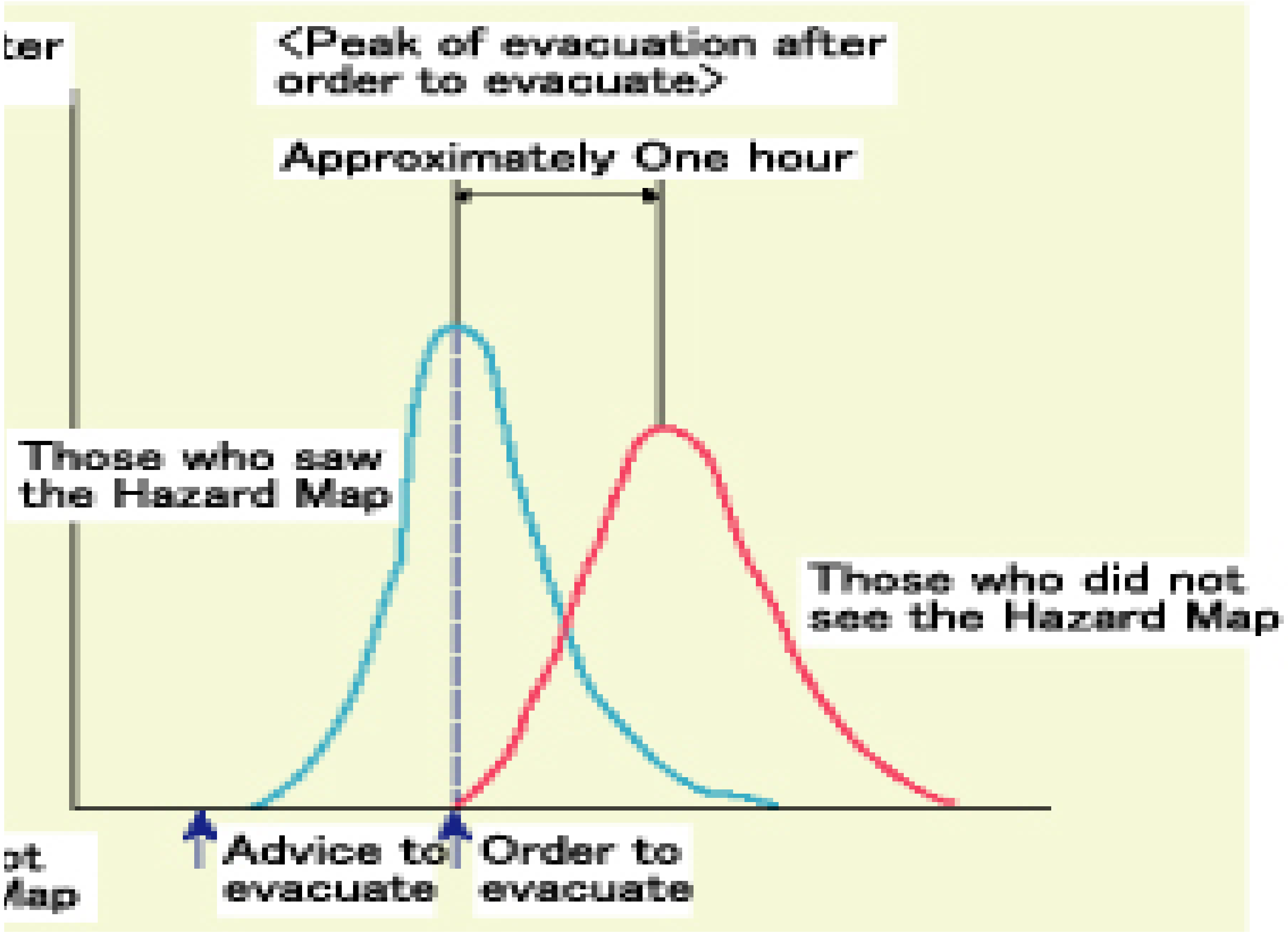
- Identify areas of potential hazards.
- Improve the abilities of agencies and individuals to respond to the consequences of a disaster event.
- Develop plans to enhance disaster response operations.
  - methods for predicting the effectiveness of disaster response operations.
  - Emergency Plans
    - Pre disaster response
    - Post disaster recovery action
- Installing early warning systems.
- Training of personnel



Preparedness Planning Cycle

Evacuation Planning

- How to evacuate citizens in the event of a major disaster in both warning and no warning situation?



Source : "Survey Report on Local Residents' Reactions in the Rainstorm in Koriyama at the End of August 1988."
Katada Laboratory, Faculty of Engineering, Gumma University

Evacuation Planning

- Advanced planning for
  - Activation procedures,
  - Determination of adequate and effective routes,
  - Methods of transportation,
  - Destinations for those evacuated,
  - Security precautions for homes and belongings,
  - Adherence by citizens to evacuation orders, and
  - Facilitating the evacuation itself.
- Full-scale test may give an accurate idea of how the plan will work in a real-life situation

Elements of Preparedness

- National Level
  - Examples: Policies, procedures, guidelines
- Local Level
  - Examples: Plans, resources, authority
- Personal Level
  - Examples: Knowledge, Skills, and Attitude

Mitigation vs. Preparedness

- *Mitigation* attempts to eliminate hazard risk by reducing either the likelihood or the consequences of the risk associated with the particular hazard
- *Preparedness*, on the other hand, seeks to improve the abilities of agencies and individuals to respond to the consequences of a disaster event <u>*once the disaster event has occurred*</u>

Mitigation vs. Preparedness

- Preparedness assumes the occurrence of an event, whereas mitigation attempts to prevent the event altogether

- Mitigation *focuses on activities taken prior to the instance of an emergency event*

- Preparedness: *look for the activities needed once a disaster has occurred*.


Response
RESPONSE
RESPONSE
RESPONSE
RESPONSE
RESPONSE


Response
Recovery
Mitigation
Preparedness

Response

- Pre- and post-disaster activities to provide emergency assistance
  - For casualties
  - To reduce the probability of secondary damages
  - To speed up recovery operations
- By the Local and Federal Government , Armed forces, and Volunteer humanitarian Groups

Emergency Operations and Relief Activities

- Evacuation & Temporary shelter
- Search and rescue
- Health care
- Food and water distribution
- Temporary sanitation
- Restoration of access to transport
- Preliminary damage assessment

Class Activity/Homework

- Keep in mind the 3 phases of DRM covered so far that are mitigation, preparedness, and response
- Now consider our university premises as your study area and
  1. Do the hazard profiling of the study area
  2. Select the hazard with the highest probability of occurrence and with the highest impact (not necessarily the same hazard qualifying both criteria)
  3. Suggest at least
     - Three (03) mitigation approaches
     - Three (03) preparedness activities
     - Three (03) response measures

DISASTER RECOVERY

UNISDR Terminologies

- **Recovery**

The restoring or improving livelihoods and health, as well as economic, physical, social, cultural, and environmental assets, systems, and activities of a disaster-affected community or society, aligning with the principles of sustainable development and "build back better," to avoid or reduce future disaster risk.

- **Rehabilitation**

The restoration of basic services and facilities for the functioning of a community or a society affected by a disaster.

Response vs. Recovery

- There is usually an overlap of disaster response and recovery work.
- Response is implemented during and in the immediate aftermath of a disaster to save lives and meet basic subsistence needs (immediate and short-term needs).
- The main objective of the recovery phase is restoring and improving people's former living conditions (in longer terms).
- Also provides a 'window of opportunity'. How???

Recovery

Rebuilding, reconstruction, and repair of the damages

- Function by which communities and individuals repair, reconstruct, or regain what has been lost as a result of a disaster.
- Recovery measures should also **reduce the risk in the future.**
- Requires special skills, equipment, resources, and personnel.
- The recovery period follows the emergency phase of a disaster.

Recovery ---conti

- Often begins in the initial hours and days following a disaster event and can continue <u>for months or years</u>, depending on the severity of the event.
- Recovery function has such long-lasting effects and usually high costs.
- Participants include all levels of government, the military, the business community, political leadership, community activists, and individuals.
- Unique to each community and depends on the extent of the damages and available resources to cope.

Actions and Activities

- Ongoing communication with the public.
- Provision of temporary houses or long-term shelter.
- Assessment of damages and needs.
- Inspection of damaged structures.
- Demolition of damaged structures.
- Clearance, removal, and disposal of debris.
- Repair of damaged structures.

Actions and Activities

- Rehabilitation of structures.
- New construction.
- Social rehabilitation programs.
- Creation of employment opportunities.
- Reimbursement for property losses.
- Rehabilitation of the injured.
- Reassessment of hazard risk.

Pre-Disaster Recovery Actions

- Hypothetical and focusing on broad goals rather than specific actions and procedures.
- Sometimes referred to as "Pre-Event Planning for Post-Event Recovery (PEPPER)."
- Can reduce the risk of haphazard rebuilding.

Pre-Disaster Recovery Actions

Examples:

- Site selection for long-term temporary houses.
- Site selection for temporary business activity.
- Site selection for disposal of debris.
- Emergency needs assessment.
- Volunteers and donation management.
- Mitigation measures and secondary hazard reduction measures.

Short-Term Recovery

- Restores vital services and systems.
- Immediate and overlaps with response.

Actions

- Restoring interrupted utility and other essential services (roads cleared).
- Reestablishing transportation routes.
- Providing food, shelter, and health care to those displaced by the incident.

Long-Term Recovery

- Restores all services to normal or better conditions.
  - both the personal lives of individuals and the livelihood of the community.
- It may involve some of the same actions as short-term ones.
- Permanent construction or replacement of severely damaged physical structures.
- Full restoration of all services and local infrastructure and the revitalization of the economy.
- May continue for a number of months or years, depending upon the severity of the damages.

Example: Complete redevelopment

Post Disaster Long-Term Recovery Planning

- Planning before the next disaster strikes.
- It provides the following benefits;
  - Identifies the most vulnerable areas of the community.
  - Accelerates approval of federal funding for rebuilding in the post-disaster environment.
  - Anticipates/compensates for regulatory and environmental requirements for rebuilding.
  - Minimizes economic and social disruption to the community.
  - Maximizes post-disaster funding in the public and private sectors.

Disasters as Opportunities for Development Initiatives

With a disaster comes disruption and tragedy,

but in the aftermath comes opportunity!

- Disasters can highlight particular areas of vulnerability.
- The political environment may favor a much higher rate of economic and social change than before in areas such as land reform, new job training, housing improvements, and restructuring of the economic base.
- Emergency lending for post-disaster investment may be used to restructure the economy as a result of a disaster.

Questions?